EFFECTIVE STERILIZED FOREIGN EXCHANGE INTERVENTION?

EVIDENCE FROM A RULE-BASED POLICY

Guido M. Kuersteiner David C. Phillips Mauricio Villamizar-Villegas

October 2026

MOTIVATION: WHY DOES THIS QUESTION MATTER?

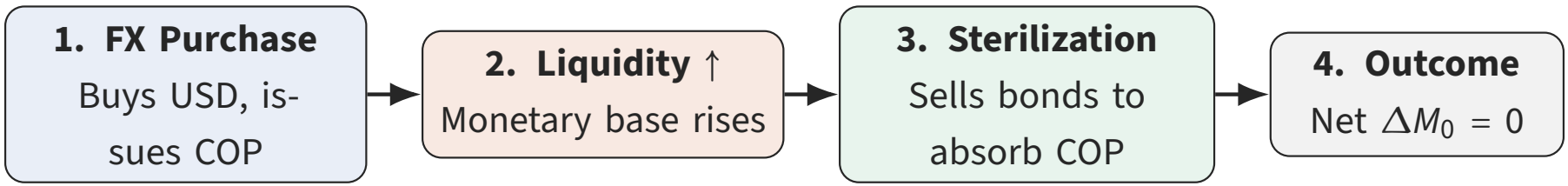
- **The Policy Problem**:
  - Central banks intervene to dampen excessive volatility that disrupts trade, inflation, and financial stability.
  - *Trilemma Conflict*: Free capital mobility and floating rates make independent exchange rate targets theoretically elusive.
- **The Empirical Obstacle in Naive Regressions**:

$$\Delta s_{t+h} = \alpha + \beta \, \text{Intervention}_t + \varepsilon_{t+h}$$

  - Interventions occur during extreme currency stress (*reverse causality*).
- Disentangling policy effectiveness from the market conditions triggering it requires an exogenous institutional threshold.

WHAT IS STERILIZED FOREIGN EXCHANGE INTERVENTION?

- Central bank buys or sells foreign currency while neutralizing domestic liquidity effects via open-market operations ($\Delta M_0 = 0$).



Channels:

- *Portfolio Balance*: Shifts relative supplies of imperfect substitute bonds.
- *Microstructure*: Absorbs dealer inventory order flow imbalances.

The Core Puzzle

If monetary policy and the monetary base are unchanged, can intervention move nominal rates in practice?

SETTING: BANCO DE LA REPÚBLICA (2001–2012)

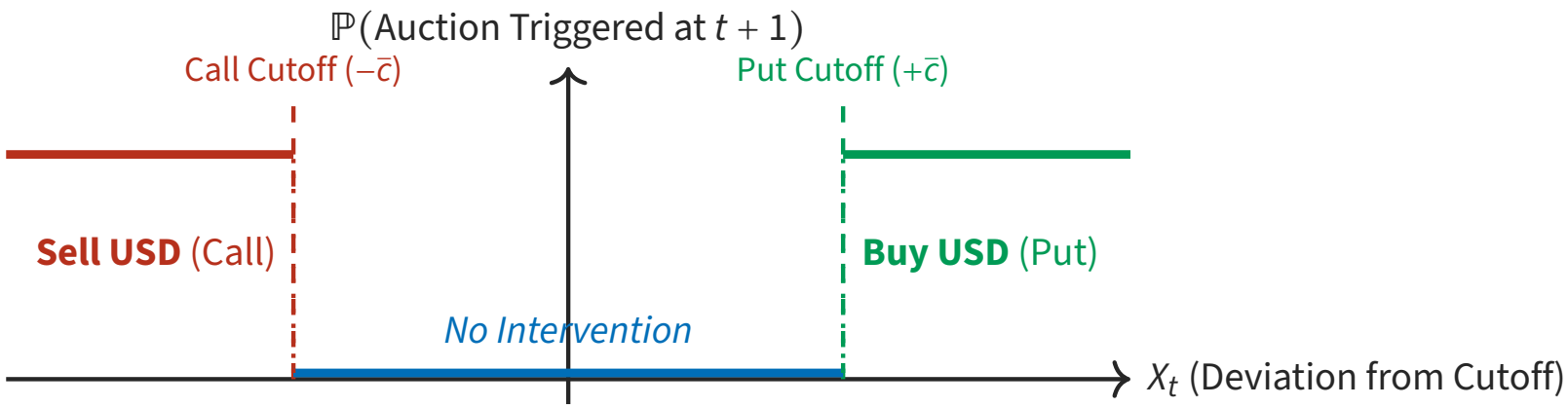
- Colombia adopted an inflation-targeting regime with a floating exchange rate in Oct 1999
- In 2002, the central bank introduced **rule-based forex interventions**
  - Trigger based on deviation from the 20-day moving average (MA_{20}).
- **Threshold Variations Across Sample**:
  - *Standard regime*: $\bar{c} = 4\%$ deviation.
  - *Tight band*: 2% (December 2005 – June 2008).
  - *Crisis band*: 5% (October 2008 – October 2009).
- The intervention program ended in Feb 2012

THE IDENTIFICATION STRATEGY

- **Construct the Running Variable**:

$$d_t \equiv \frac{e_t - \bar{e}_t}{\bar{e}_t} \times 100\%, \qquad X_t \equiv d_t - r_t$$

where e_t is the closing rate, $\bar{e}_t$ is the 20-day MA, and r_t is the codified threshold ($\pm 4\%$). Trigger occurs when $X_t \geq 0$.



IDENTIFICATION: ASSIGNMENT AND THE LOCAL CONTRAST

- **Treatment Assignment**:
  - *Treated*: Exchange-rate deviation crosses the cutoff ($X_t \geq 0$).
  - *Control*: Deviation falls just short of the cutoff ($X_t < 0$).
  - *Treatment of Interest*: Rule-triggered auction of forex options.
- **Compare observations near the cutoff**:
  - Compare days where the threshold was **barely crossed** vs. **barely missed**: $X_t \in [-\epsilon, 0)$ vs. $X_t \in [0, +\epsilon]$
  - Local equivalence: Days within a narrow bandwidth share identical macroeconomic conditions, sentiment, and trend momentum.

KEY IDENTIFICATION ASSUMPTIONS

- **Quasi-Random Assignment Around Market Close**:
  - e_t is a volume-weighted average of trades up to 4:00 PM.
  - Unpredictable minute-level trading noise η_t rules out precise sorting:

$$X_t = x_t^* + \eta_t, \quad \eta_t \mid x_t^* \sim F_\eta \quad \text{(smooth density)}$$

  - Assignment in the immediate neighborhood of $X_t = 0$ is locally randomized.
- **Continuity of Potential Outcomes**:

$$\lim_{x \downarrow 0} \mathbb{E}[Y_t(0) \mid X_t = x] = \lim_{x \uparrow 0} \mathbb{E}[Y_t(0) \mid X_t = x]$$

- In the counterfactual, expected returns do not jump. McCrary test shows no density jump at cutoff ($p > 0.40$); pre-intervention covariates (VIX, EMBI spreads, oil prices) are balanced.

DATA

Component	Empirical Description
Country & Sample	Colombia: Jan 2002 – Feb 2012 (entire program)
Exchange Rates	High-frequency tick-by-tick FX transactions (≈ 3.5M trades)
Interventions	Auction announcements and option exercises
Financial Data	Bank balance sheets & interest rates
Capital Flows	Daily portfolio inflows and outflows

EMPIRICAL SPECIFICATION: DYNAMIC LOCAL LINEAR RD

Local Projections RD Specification at Horizon h

$$Y_{t+h} = \alpha_h + \tau_h D_t + \beta_- X_t + \beta_+ (D_t \times X_t) + \varepsilon_{t+h}$$

- $Y_{t+h} \equiv \ln S_{t+h} - \ln S_{t-1}$: Cumulative exchange-rate return from $t-1$ to $t+h$
- $D_t \equiv \mathbf{1}(X_t \geq 0)$: Treatment indicator
- τ_h: Discontinuity parameter capturing the causal jump at horizon h.
- β_-, β_+: Local linear slopes below and above the cutoff.
- Imbens-Kalyanaraman (2012) optimal bandwidth selection (h_{IK}).
- Use Newey-West HAC standard errors

FINDING 1: INTERVENTIONS ARE EFFECTIVE BUT TRANSITORY

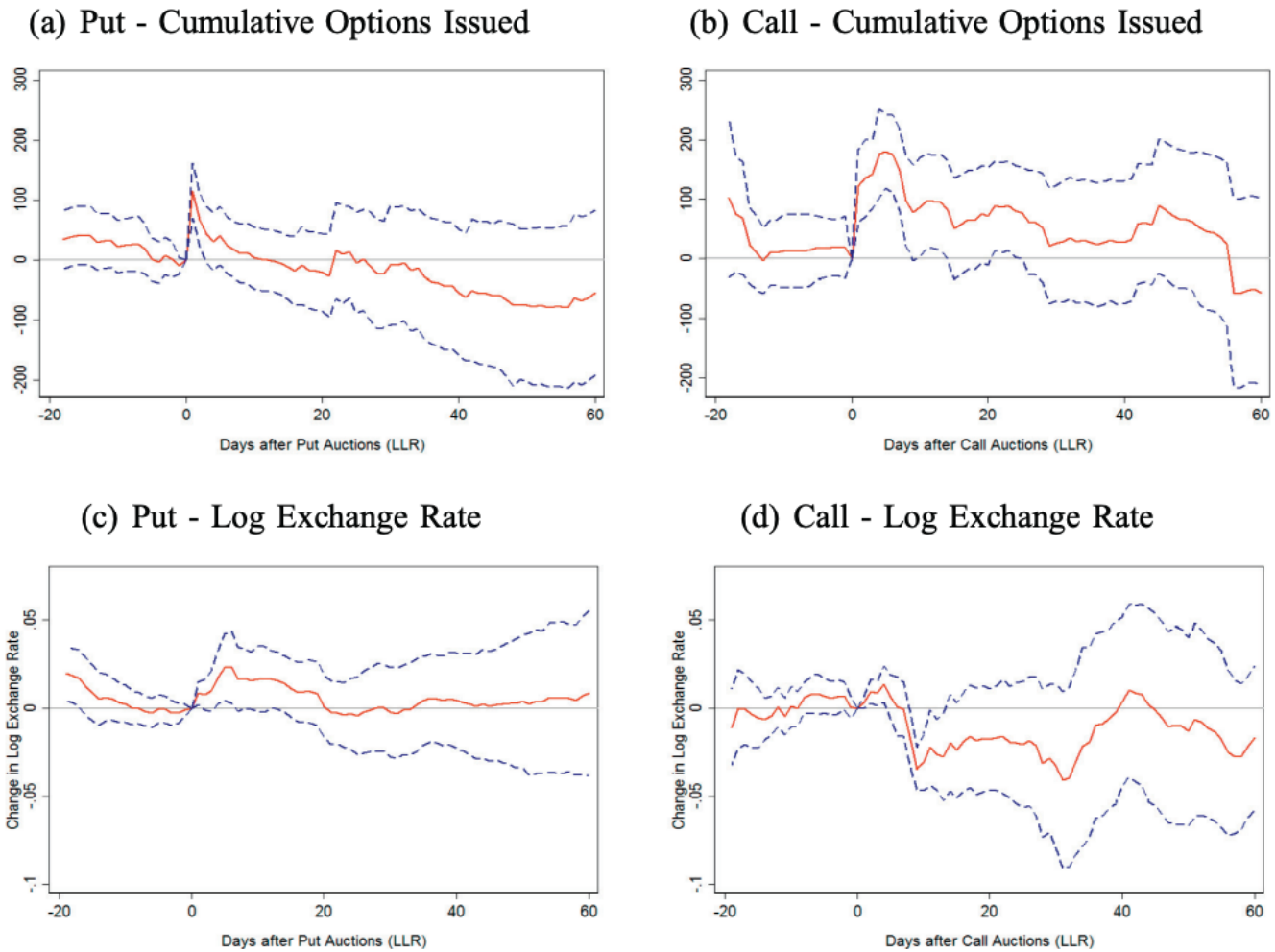


Fig. 5. IRF of effects of triggering the rule, 60 days. In sub-figures (a) and (b) the dependent variable is the volume of options issued, in millions of dollars, between the date in question and the auction. In sub-figures (c) and (d) the dependent variable is the log daily average spot rate relative to the log average spot rate the day before the auction. The solid curves present a series of regression discontinuity estimates implemented using local linear regression of the dependent variable on the running variable using daily data. Dashed curves display 95% confidence intervals of the estimates.

FINDING 1: INTERVENTIONS ARE EFFECTIVE BUT TRANSITORY

- **Dollar Purchases (Puts)**:
  - Spot rate jumps $\approx 0.8\%$ on auction day ($p < 0.01$).
  - Effect peaks at $\approx 2.0\%$ one week out, then fully reverts within 2–3 weeks.
- **Dollar Sales (Calls)**: Effect is small and asymmetric; the peso appreciates only modestly.
- **Sterilization Confirmed**:
  - Interbank interest rates show no significant response $\Longrightarrow$ not monetary policy in disguise.
- **Capital Controls**:
  - Effects appear even under free capital mobility, but controls significantly amplify the impact.

FINDING 2: EFFECT AMPLIFY UNDER CAPITAL CONTROLS

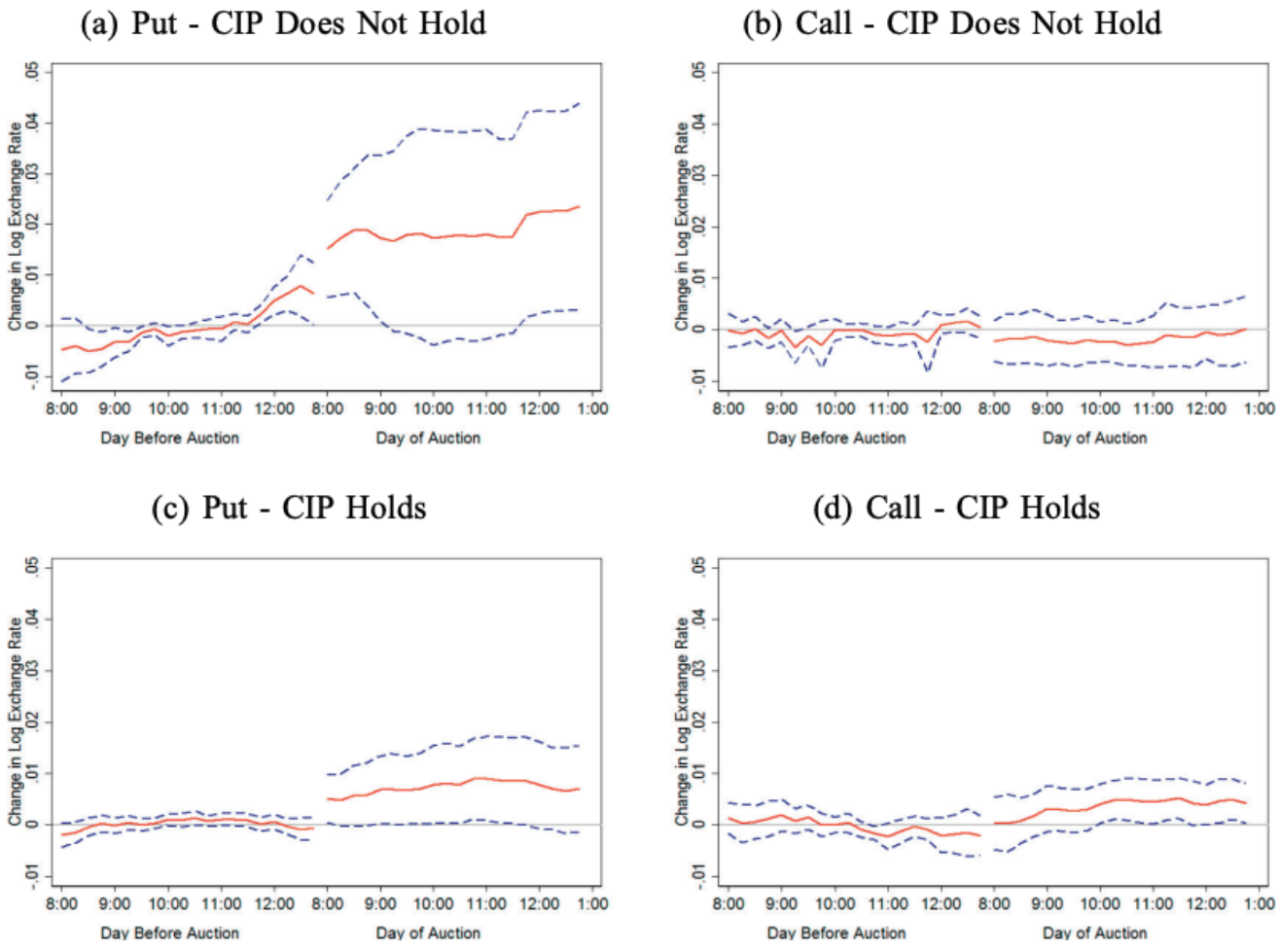


Fig. 6. IRF of log exchange rate depending on the CIP condition: $+/-$ 1 day. The dependent variable is the log average spot rate in 15-min increments relative to the log average spot rate the day before the auction. The solid curve presents a series of regression discontinuity estimates implemented using local linear regression of the dependent variable on the running variable. Dashed curves display 95% confidence intervals of the estimates. We define the time CIP holds as July 1, 2003 to June 30, 2008 and the time CIP does not hold as all dates before and after this time period.

THE CHANNELS BEHIND THE EFFECT

- **Portfolio Balance Channel (Temporarily Active)**:
  - Banks reduce USD spot holdings in the short run (Days 1–20), then adjust their positions through forward contracts by Day 60.
  - This suggests that the intervention temporarily affects banks' spot-market positions, while forward hedging may help them manage the associated currency risk.
- **Delayed Global Capital Flows**:
  - Foreign investors adjust little in the first 10 days (≈ $5M), then accelerate to large cumulative outflows of ≈ $100M by day 60, returning the exchange rate to trend.

LIMITATIONS & IDENTIFICATION CAVEATS

- **Anticipation & Timing**
  - Markets may anticipate intervention before the cutoff; 61% of the estimated jump comes from control days moving down.
  - RD may capture intervention happening sooner, rather than its long-run effect.
- **Dynamic Effects**
  - Estimated effects combine the initial intervention with future interventions triggered by the policy rule.
- **Data Limitations**
  - Results for dollar sales are noisy; cover only 5 domestic banks.
  - CIP violations are an imperfect proxy for capital controls.
- **Generalizability**
  - Results are local to the cutoff and may not generalize to other countries or intervention policies.